QUAID-I-AZAM UNIVERSITY ISLAMABAD

DEPARTMENT OF COMPUTER SCIENCE



DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

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Submitted to:

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Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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Chapter No. 01

Software Project Management Plan

3.1.3 Implementation Phase

3.1.3.1 System Implementation

Task Description	This phase makes a speciality of the improvement and integration of the clickbait detection device with the aid of applying device gaining knowledge of techniques, textual content evaluation techniques, and machine additives to provide prediction and rationalization effects.	
Deliverables	After completing the software requirements, use case details and diagrams will be created.	
Resources Needed	**People**	Faizan Ahmad, Zohaib Zafar
	Software	Microsoft Word, Python, Machine Learning libraries, Web development Tools
	Hardware	Laptop
Dependencies and Constraints	We have prepared clickbait statistics and trained models that may be used in the mission.	
Risks	Model consequences depend upon the exceptional and balance of the schooling facts.	

3.2 Assignments

Task	Assignee	Role
Requirement identification	FAIZAN AHMAD, ZOHAIB ZAFAR	SYSTEM Analyst
Defining Use Cases		SYSTEM Analyst
Develop Domain Model		SYSTEM Analyst, Software Architect.
ERD Design		Database Designer, SYSTEM Analyst
Testing and Validation		Performing device testing to confirm features and outcomes.
Documentation		Creating complete technical documentation for the very last year assignment.
User Interface Design		Designing how customers engage with the system and notice consequences.
Class Diagram Designing		Software Architect, SYSTEM Analyst.

Sequence Diagrams		SYSTEM Analyst, Software Architect
Design Phase Verification		Quality Assurance Engineer, SYSTEM Software Architect.
Software Test Documentation		Quality Assurance Engineer, Test Analyst
Implementation Phase		Software Developer.
Review and Evaluation		Reviewing challenge development, validating outcomes, and providing instructional steerage.

3.3 Project Schedule

		Name	Duration	Start	Finish	Predecessors	Resource Names
1		**Project Initiation & Pla...**	**11 days?**	**11/17/25, 8:00AM**	**12/1/25, 5:00PM**		
2		Topic finalization	2 days?	11/17/25, 8:00AM	11/18/25, 5:00PM		
3		Problem statement defini...	2 days?	11/19/25, 8:00AM	11/20/25, 5:00PM	2	
4		Literature review (existi...	1 day?	11/21/25, 8:00AM	11/21/25, 5:00PM	3	
5		Project scope & objectiv...	3 days?	11/24/25, 8:00AM	11/26/25, 5:00PM	4	
6		Initial project plan approva	3 days?	11/27/25, 8:00AM	12/1/25, 5:00PM	5	
7		**Requirement Analysis**	**9 days?**	**11/28/25, 8:00AM**	**12/10/25, 5:00PM**		
8		Functional requirements ...	2 days?	11/28/25, 8:00AM	12/1/25, 5:00PM		
9		Non-functional requirem...	2 days?	12/2/25, 8:00AM	12/3/25, 5:00PM	8	
10		Use case identification (...	2 days?	12/4/25, 8:00AM	12/5/25, 5:00PM		
11		Requirements validation	3 days?	12/7/25, 8:00AM	12/10/25, 5:00PM		
12		Requirements document...	2 days?	12/9/25, 8:00AM	12/10/25, 5:00PM		
13		**System Design**	**17 days?**	**12/11/25, 8:00AM**	**1/2/26, 5:00PM**		
14		High-level system archit...	2 days?	12/11/25, 8:00AM	12/12/25, 5:00PM		
15		Use Case Diagram design	3 days?	12/13/25, 8:00AM	12/17/25, 5:00PM		
16		System Sequence Diagra...	3 days?	12/18/25, 8:00AM	12/22/25, 5:00PM	15	
17		Sequence Diagrams (UC-...	4 days?	12/23/25, 8:00AM	12/26/25, 5:00PM	16	
18		Database & data flow de...	3 days?	12/24/25, 8:00AM	12/26/25, 5:00PM		
19		Design review & refinem...	5 days?	12/29/25, 8:00AM	1/2/26, 5:00PM	18	
20		**Model Development**	**34 days?**	**1/20/26, 8:00AM**	**3/6/26, 5:00PM**		
21		Dataset collection	4 days?	1/20/26, 8:00AM	1/23/26, 5:00PM		
22		Data cleaning & preproc...	5 days?	1/26/26, 8:00AM	1/30/26, 5:00PM	21	
23		Feature extraction & sel...	5 days?	2/2/26, 8:00AM	2/6/26, 5:00PM		
24		Machine learning model s..	5 days?	2/9/26, 8:00AM	2/13/26, 5:00PM		
25		Model training & evaluatior	15 days?	2/16/26, 8:00AM	3/6/26, 5:00PM	24	
26		**System Development**	**34 days?**	**3/2/26, 8:00AM**	**4/16/26, 5:00PM**		
27		User authentication module	5 days?	3/2/26, 8:00AM	3/6/26, 5:00PM		
28		Headline input & predicti...	5 days?	3/9/26, 8:00AM	3/13/26, 5:00PM	27	
29		Explanation generation ...	6 days?	3/16/26, 8:00AM	3/23/26, 5:00PM	28	
30		Detection history manag...	5 days?	3/21/26, 8:00AM	3/27/26, 5:00PM		
31		Model performance visu...	5 days?	3/30/26, 8:00AM	4/3/26, 5:00PM	30	
32		system integration	9 days?	4/4/26, 8:00AM	4/16/26, 5:00PM		
33		**Testing & Validation**	**17 days?**	**4/16/26, 8:00AM**	**5/8/26, 5:00PM**		
34		Unit testing	3 days?	4/16/26, 8:00AM	4/20/26, 5:00PM		
35		Functional testing	8 days?	4/21/26, 8:00AM	4/30/26, 5:00PM	34	
36		Acceptance testing	3 days?	5/1/26, 8:00AM	5/5/26, 5:00PM	35	
37		Bug fixing & validation	3 days?	5/6/26, 8:00AM	5/8/26, 5:00PM	36	
38		**Documentation**	**13 days?**	**5/11/26, 8:00AM**	**5/27/26, 5:00PM**		
39		Technical documentation...	6 days?	5/11/26, 8:00AM	5/18/26, 5:00PM		
40		Diagram & table updates	4 days?	5/19/26, 8:00AM	5/22/26, 5:00PM	39	
41		Proofreading & formatting	3 days?	5/25/26, 8:00AM	5/27/26, 5:00PM	40	
42		**Final Review & Submis...**	**7 days?**	**5/26/26, 8:00AM**	**6/3/26, 5:00PM**		
43		Supervisor review	3 days?	5/26/26, 8:00AM	5/28/26, 5:00PM		
44		Final corrections	2 days?	5/29/26, 8:00AM	6/1/26, 5:00PM	43	
45		Final submission	2 days?	6/2/26, 8:00AM	6/3/26, 5:00PM	44	

3.4 Gantt Chart

